

Day 1 Mastery Quiz — Instructor Key

Visible Output and Stored State

Display-only value → printed stored value → separate report value → later overwrite

Instructor key. Same layout as student copy; solutions filled in response boxes. Print format: duplex, left-stapled packet.

Name		Section		Date	
Score	_____/10		Mastery check		secure / developing / needs support

A ramp-gate team checks whether the notebook record supports the value that will go into a lab table. A value can be stored, printed, displayed as a final expression, or overwritten later. Your trace must say which value is stored **after each cell**; visible output alone is not enough evidence.

1 Reference code for Items 1–10

[predict](#) [trace](#)

```
1 # Cell A: first gate check
2 run_id = "gate-1"
3 distance_m = 10.0
4 time_s = 4.0
5 speed_mps = distance_m / time_s
6 print("raw", speed_mps)
7 speed_mps
8
9 # Cell B: second gate check with a candidate correction
10 run_id = "gate-2"
11 time_s = 2.0
12 speed_mps = distance_m / time_s
13 correction_mps = 0.5
14 speed_mps + correction_mps
15 print("stored after candidate", speed_mps)
16
17 # Cell C: store the reviewed report value separately
18 reported_speed = speed_mps + correction_mps
19 print("reported", reported_speed)
20
21 # Cell D: record, then make one more candidate update
22 speed_mps = reported_speed
23 reported_speed = reported_speed + 0.2
24 print("final stored", speed_mps)
25 print("latest candidate", reported_speed)
```

2 Items 1–2: recognize the evidence roles

[recognize](#) [predict](#)

Pt.	Prompt	My response
1	Which variables store physical measurement inputs from the ramp-gate run, not labels or calculated/corrected results?	Key: distance_m and time_s.
1	Which exact line first makes 5.5 visible without storing it in either speed_mps or reported_speed?	Key: speed_mps + correction_mps in Cell B.

3 Items 3–5: trace the stored values after later cells

[trace](#) [explain](#)

For a value that does not exist yet, write **not defined**. Name both visible evidence and stored state when they differ.

Pt.	Checkpoint	Stored speed_mps	Stored reported_speed	Visible evidence that could mislead a record
1	After Cell B	Key: 5.0	Key: not defined	Key: Displayed 5.5; printed stored after candidate 5.0.
1	After Cell C	Key: 5.0	Key: 5.5	Key: Printed reported 5.5; speed_mps not overwritten yet.
1	After Cell D	Key: 5.5	Key: 5.7	Key: Printed final stored 5.5; latest candidate 5.7.

4 Items 6–8: debug two tempting record claims

debug test

Claims to check

Claim 1, after Cell B: The recorded speed is 5.5 because the notebook showed 5.5.

Claim 2, after Cell D: The final stored `speed_mps` is 5.7 because the notebook printed `latest candidate` 5.7.

Pt.	Prompt	My response
1	What is wrong with Claim 1? Use stored/displayed language and give the stored value after Cell B.	Key: 5.5 is display-only after Cell B. Stored <code>speed_mps</code> is 5.0; <code>reported_speed</code> is not defined.
1	What is wrong with Claim 2? Use the order of the two assignments in Cell D and give the final stored <code>speed_mps</code> .	Key: Cell D first copies <code>reported_speed</code> into <code>speed_mps</code> , so <code>speed_mps</code> becomes 5.5; later only <code>reported_speed</code> becomes 5.7.
1	Write one Python line you could run after Cell D to check <code>speed_mps</code> without changing any variable.	Key: <code>print(speed_mps)</code> or bare <code>speed_mps</code> .

5 Item 9: complete the safer edit

implement trace

Complete the line so Cell C stores the reviewed report value separately and does not overwrite `speed_mps` yet.

```
----- = speed_mps + correction_mps
print("reported", reported_speed)
```

Pt.	Prompt	My response
1	Write the completed line or the missing variable name.	Key: <code>reported_speed = speed_mps + correction_mps</code>

6 Item 10: transfer the evidence rule

transfer explain

Pt.	Prompt	My response
1	In one sentence, explain why the two printed Cell D lines are useful evidence instead of just printing the latest candidate value.	Key: They separate the recorded stored value from the later candidate; otherwise 5.7 could be mistaken for current <code>speed_mps</code> .

Support route, not scored

My issue is mostly: setup / Python syntax / tracing / arithmetic / notebook display / assignment order / confidence / time.

My next small action is: rerun one cell / rebuild the trace table / ask a partner / ask instructor / try a recovery version.